

5. Error Analysis Note

Submitted detector: OpenCLIP ViT-H/14 spatial model. Errors are counted on the 5,000-image WildFake val slice (2,500 real / 2,500 AI) at threshold 0.5. Fake = AI-generated. A false positive flags a real photo as AI; a false negative lets an AI image through as real.

6 False positives (clean)	298 False negatives (clean)	0.997 Precision @ 0.5	0.881 Recall @ 0.5
-------------------------------------	---------------------------------------	---------------------------------	------------------------------

False positives - rare, high-confidence when they happen

On clean images the model almost never accuses a real photo: 6 FP out of 2,500 reals (0.24%). Those six are not borderline - the strongest has $P(\text{AI}) = 0.91$. CLIP's global semantics can still read a real scene as generated when lighting, texture, or composition looks synthetic.

Heavy resharing inflates this class. Gaussian noise at 10% raises FP to 64; blur sigma=2.0 raises it to 43. Additive grain and strong blur are the transforms that make real photos look least photographic to the tower.

False negatives - the dominant error

298 of 2,500 AI images are missed on clean (11.9%). The worst misses are not near the threshold: several score $P(\text{AI}) \sim 0$. Photorealistic generators (later diffusion / Midjourney-style images in WildFake) land on the same semantic manifold as real photos, and a spatial CLIP tower has little local forensic signal to fall back on.

Noise and blur also drive FN up (437 and 429). That matches the official grid: Noise 10% is the worst cell (AUC 0.9707). The model still ranks well; at 0.5 it simply stays on the call-it-real side of the logit.

Where errors grow

Condition	FP	FN	Precision	Recall
Clean	6	298	0.9973	0.8808
Color	11	302	0.9950	0.8792
JPEG 30	10	360	0.9953	0.8560
Crop 80%	16	352	0.9926	0.8592
Resize 1/4	21	365	0.9903	0.8540
Blur 2.0	43	429	0.9797	0.8284
Noise 10%	64	437	0.9699	0.8252

Same 5k slice, threshold 0.5. JPEG / color / crop barely move FP. Noise and strong blur move both FP and FN.

Trade-offs in the proposed approach

Threshold 0.5 is conservative on purpose

Precision 0.997 vs recall 0.881. For a platform, accusing a real photo is worse than missing some AI. Lowering the threshold would cut FN and raise FP; we did not retune it on this val slice.

- Spatial-only vs a forensic second tower.

- Ranking vs operating point.

RGPA ma
and loses
at the cos

Official AI
order; the

- **Robustness training vs extra eval enhancement.**

Single-de
small clea

- **Capacity vs speed.**

Truncating
is what ho